

Section 28: ACLs

Layer: L3/L4 | Reference: Odom Vol2 Ch6-7: Standard and Extended ACLs | *Original study notes — CCNA 200-301*

Access Control Lists (ACLs) filter traffic based on Layer 3 and Layer 4 criteria. They are processed top-down with an implicit deny at the end, and are one of the most heavily tested configuration topics on the CCNA exam.

Key Concepts

- Standard ACLs filter on source IP address only (numbered range 1-99, 1300-1999)
- Extended ACLs filter on source IP, destination IP, protocol, and port numbers (numbered range 100-199, 2000-2699)
- Named ACLs can be standard or extended and allow individual line editing without rewriting the whole list
- Processing is top-down, first-match wins; once a line matches, processing stops
- Every ACL ends with an implicit 'deny any' - if nothing matches, traffic is dropped
- One ACL per protocol, per direction, per interface (e.g., one inbound IPv4 ACL and one outbound IPv4 ACL on the same interface, but not two inbound)
- Standard ACLs should be placed as close to the destination as possible, since they can't filter on destination, placing them near the source would block traffic to ALL destinations
- Extended ACLs should be placed as close to the source as possible, since they're specific enough not to over-filter
- Wildcard masks are inverted logic from subnet masks: 0 = bit must match, 1 = bit is ignored
- 'host 10.1.1.1' is shorthand for '10.1.1.1 0.0.0.0'; 'any' is shorthand for '0.0.0.0 255.255.255.255'
- Apply direction is relative to the router: 'in' = traffic entering the interface from outside, 'out' = traffic leaving the interface
- VTY line ACLs use 'access-class' instead of 'ip access-group' since they're not bound to a physical interface

Command Reference

Command	Purpose
<code>access-list <1-99> {permit deny} <src> <wildcard></code>	Numbered standard ACL entry
<code>access-list <100-199> {permit deny} <proto> <src> <wm> <dst> <wm> [eq port]</code>	Numbered extended ACL entry
<code>ip access-list standard <name></code>	Begin a named standard ACL (enters sub-config mode)
<code>ip access-list extended <name></code>	Begin a named extended ACL (enters sub-config mode)
<code>ip access-group <name number> {in out}</code>	Apply an ACL to an interface in a given direction
<code>access-class <name number> {in out}</code>	Apply an ACL to VTY (Telnet/SSH) lines

Command	Purpose
<code>show access-lists</code>	Display all configured ACLs and their match counters
<code>show ip interface <intf></code>	Confirm which ACLs are applied to an interface and in which direction

Configuration Walkthrough

Standard ACL: permit only one host, deny all others, applied near destination

```
R1(config)#access-list 10 permit host 10.1.1.50
R1(config)#access-list 10 deny any
R1(config)#interface GigabitEthernet0/1
R1(config-if)#ip access-group 10 out
```

Extended named ACL: block Telnet from a subnet, permit everything else

```
R1(config)#ip access-list extended BLOCK_TELNET
R1(config-ext-nacl)#deny tcp 10.10.10.0 0.0.0.255 any eq 23
R1(config-ext-nacl)#permit ip any any
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip access-group BLOCK_TELNET in
```

Restrict VTY access to a management subnet only

```
R1(config)#access-list 5 permit 10.10.99.0 0.0.0.255
R1(config)#line vty 0 4
R1(config-line)#access-class 5 in
```

Worked Scenario

Q: You write an extended ACL to block one host from reaching a web server, but legitimate users also lose access. What's the likely cause?

Almost always a missing explicit permit before the implicit deny. If the ACL has a deny statement for the bad host and no matching permit statement for everyone else, the implicit 'deny any' at the end blocks all remaining traffic, including legitimate users. Always end a custom ACL with an explicit 'permit ip any any' (or similarly scoped permit) unless you genuinely intend a default-deny posture.

Common Mistakes / Gotchas

- Placing standard ACLs near the source, which over-blocks traffic to unintended destinations
- Forgetting the implicit deny any at the end of every ACL
- Writing the wildcard mask as if it were a subnet mask (using 255.255.255.0 instead of 0.0.0.255)
- Applying the ACL in the wrong direction (in vs out) relative to the interface
- Editing a numbered ACL line-by-line in older IOS without realizing inserting lines requires sequence numbers in newer IOS via the named ACL syntax

Exam Tips

- ★ *Standard = source only, near destination. Extended = source+dest+port, near source. This placement rule is tested directly.*
- ★ *Practice converting subnet masks to wildcard masks quickly: wildcard = 255.255.255.255 minus the subnet mask*
- ★ *Remember VTY lines use access-class, not ip access-group*